

Risk Assessment of Insurance Policies using Machine Learning Techniques

Lukáš Trstenský, Bartosz Kochanski and Maia Ghionea

<https://github.com/SAKULAK/ITU-MachineLearning-FinalProject-Couriers>

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1 Introduction

This paper aims to explore methods to model the claims risk of insurance policies using various machine learning methods as well as discuss their implementation. We will explore and clean the data, then define the claims risk based on the data available in the dataset.

1.1 Machine Learning Methods

In this paper we will use custom implementation of a Feed-Forward Neural Network Regressor and custom implementation of a Decision Tree Regressor using only Python numerical libraries, e.g. Python. Furthermore, we will use an ensemble method using a combination of TBD

2 Data

The dataset that we used for this report is the French Motor Third-Part Liability (`freMTPL2freq`) dataset from the `CASdatasets` (Dutang et al., 2024, p. 71) R package.

2.1 Data Format

See Table 1 for names and descriptions of the data columns.

2.2 Data Cleaning

For data cleaning we decided to take a conservative approach minimally modify the data. We concluded that the data did not have any obvious signs of subverting our assumptions, except for the following:

- 1. Exposure should be in range (0, 1] since the data was collected over one year, so we assume that any value above one is an error, and we set it equal to the max, so 1.
- 2. We removed any data points where *VehAge* > 30 as this is the cutoff at which Vehicles become classified as historic cars, and are treated differently.

2.3 Visualization

yappity

3 Features

describe why features r important/what they r used for here etc etc

Table 1: Data columns and descriptions.

Name	Description
IDpol	Unique policy ID.
ClaimNb	Number of claims made in policy.
Exposure	Duration of policy in years.
Area	Density category of driver's residence from "A" for rural area to "F" for urban centre.
VehPower	Power category of car.
VehAge	Age of car.
DrivAge	Age of driver.
BonusMalus	Bonus/malus, between 50 and 350: < 100 means bonus, > 100 means malus.
VehBrand	The car brand divided in the following groups: A- Renault Nissan and Citroen, B- Volkswagen, Audi, Skoda and Seat, C- Opel, General Motors and Ford, D- Fiat, E- Mercedes Chryslerand BMW, F- Japanese (except Nissan) and Korean, G- other.
VehGas	Type of gas that the vehicle uses.
Density	Number of inhabitants per <i>km</i> ² in the city the driver of the car lives in.
Region	The policy region in France (based on the 1970-2015 classification).

3.1 Parameters

which and why, maybe use itemize and make it as a list

3.2 Clustering

yuh yuh yap

3.3 PCA

yap yap yap

4 Method 1

,

5 Method 2

6 Method 3

7 Limitations

7.1 Limiting Factors for feature extraction

7.2 Limiting Factors for Performance

8 Conclusion

8.1 Improvements

braaa crrkk

References

[Dutang et al., 2024] Dutang, C., Charpentier, A., Gallic, E., and Siharath, J. (2024). CASdatasets: Insurance datasets. <https://cas.uqam.ca/pub/web/CASdatasets-manual.pdf>. R package version 1.2-0.